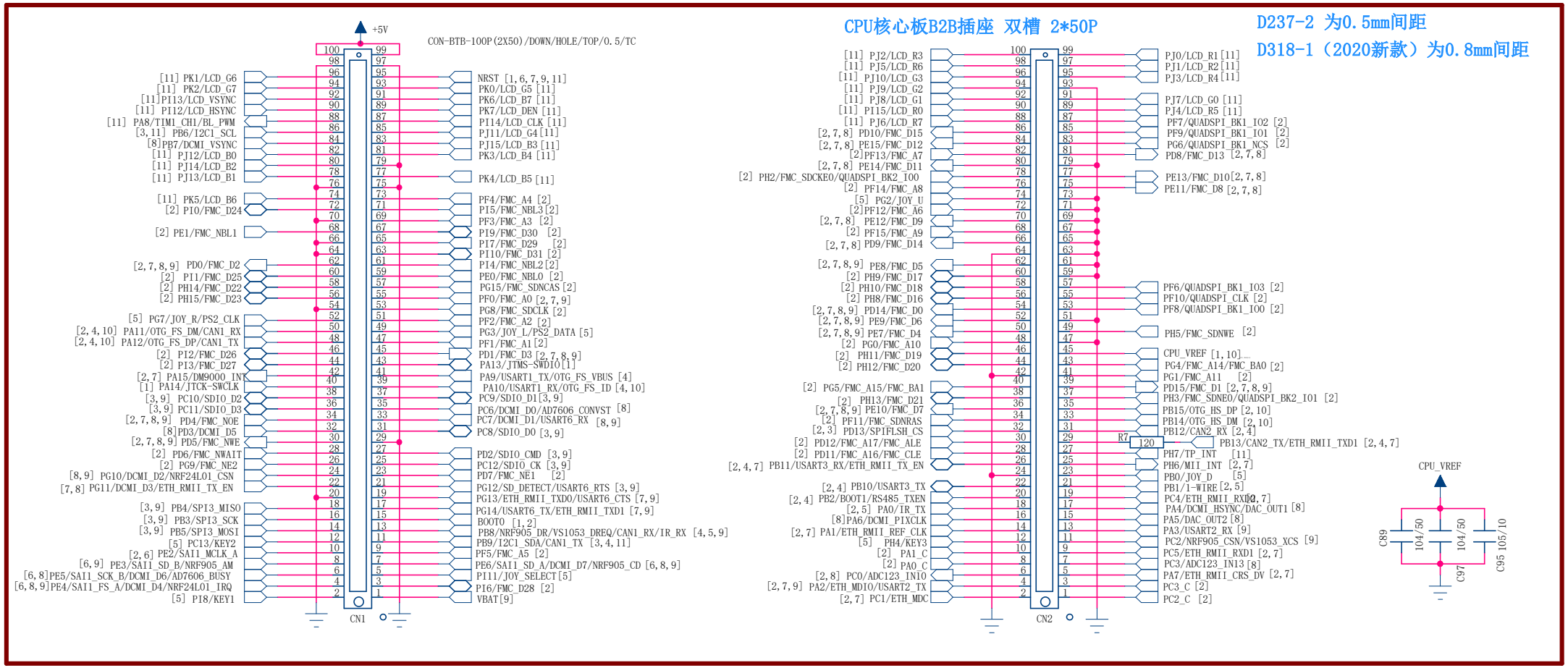
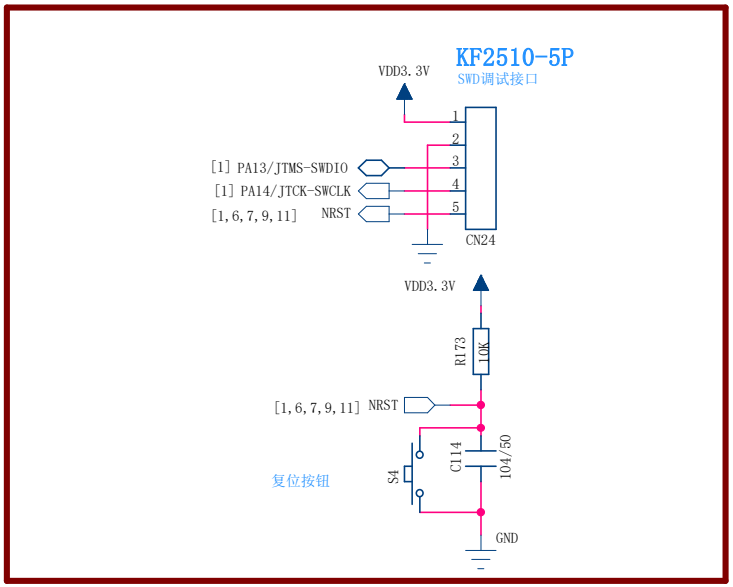




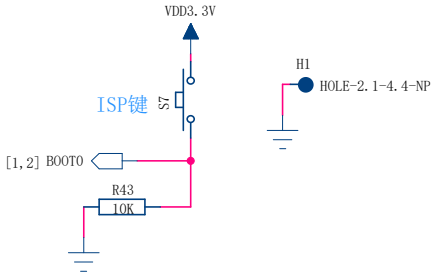
启动模式

BOOT0 = 0 BOOT1 = X 从FLASH中启动

BOOT0 = 1 BOOT1 = 0 从系统代码启动

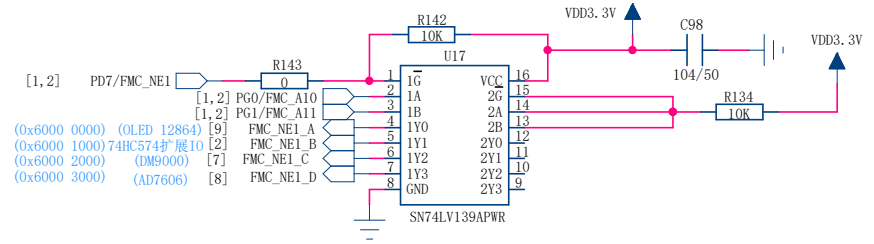
先按住ISP按钮，再按一下复位键

CPU进入ISP状态（即系统代码启动）

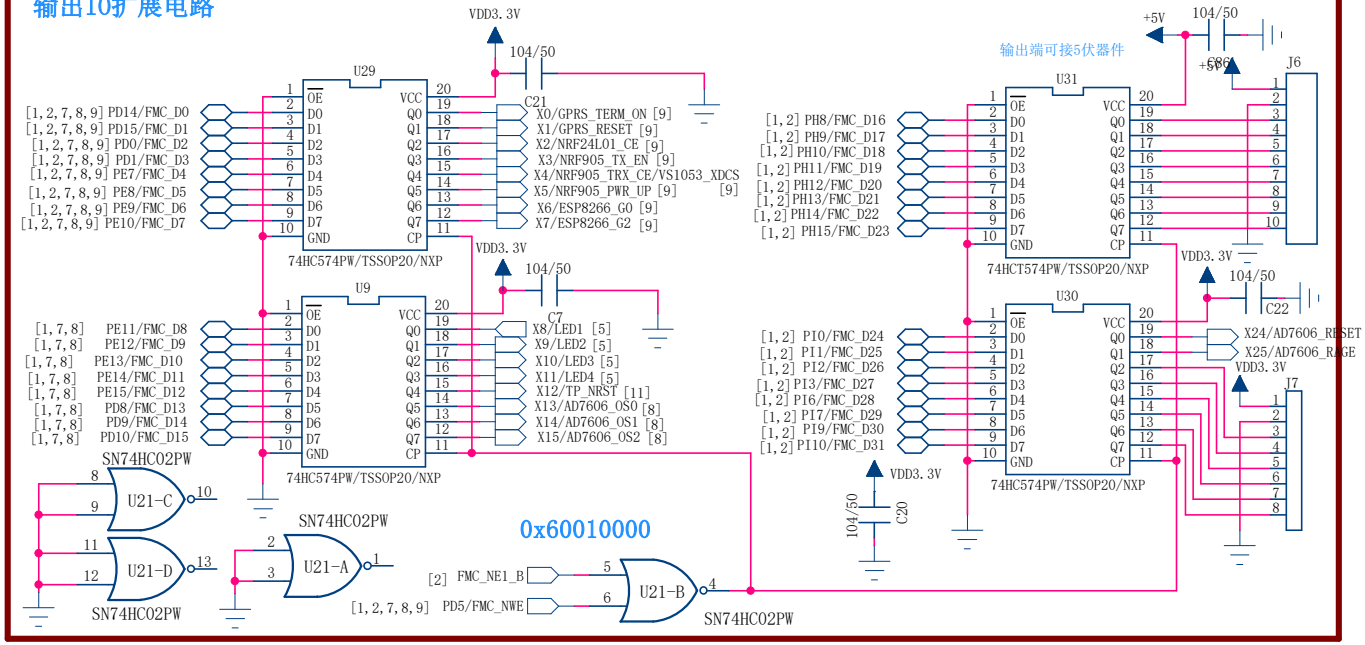


COMPANY: 安富莱电子 www.armfly.com		
TITLE: 【安富莱】STM32-V7开发板主板 D237-2 （D318-1原理图相同）		
SIZE: A3	SHEET NAME: [1]CPU	REV: 1.0
DRAWN BY: armfly	DATE: 2021-01-04	SHEET: 1 OF 11

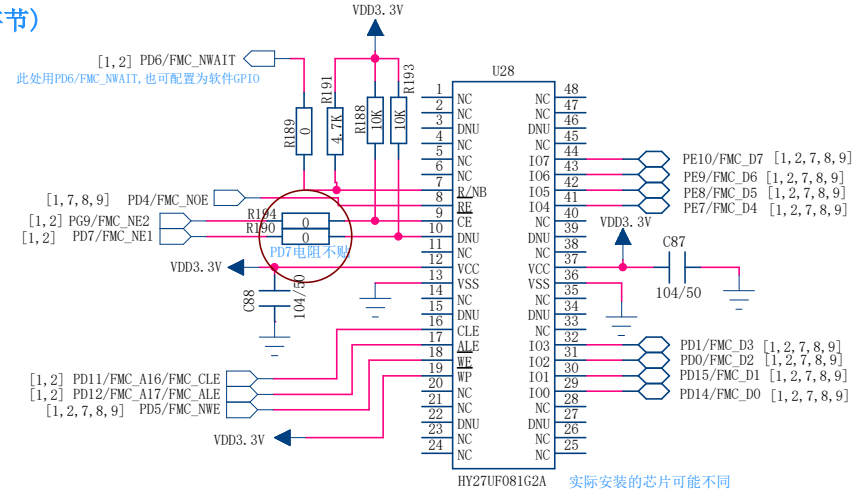
地址译码器



输出IO扩展电路



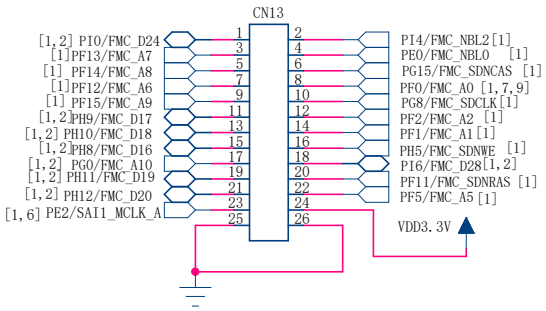
NAND FLASH(128M字节)
0x80000000



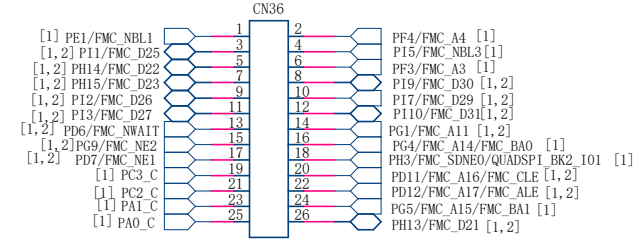
网标左边或右边[]内的数字表示该网络存在的页号，这是设计软件自动添加的

GPIO引出排针

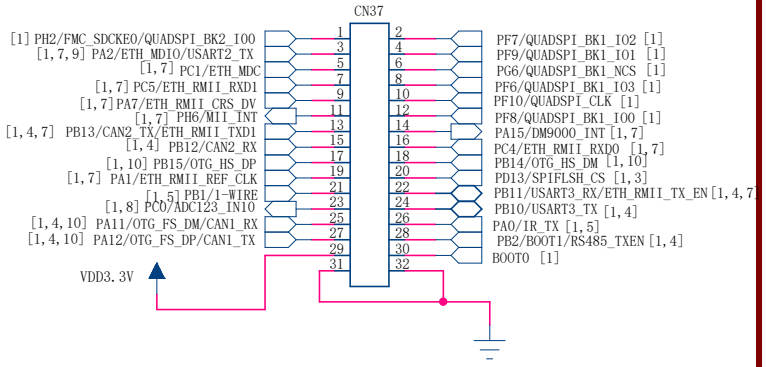
2*13P 2.54mm双排针



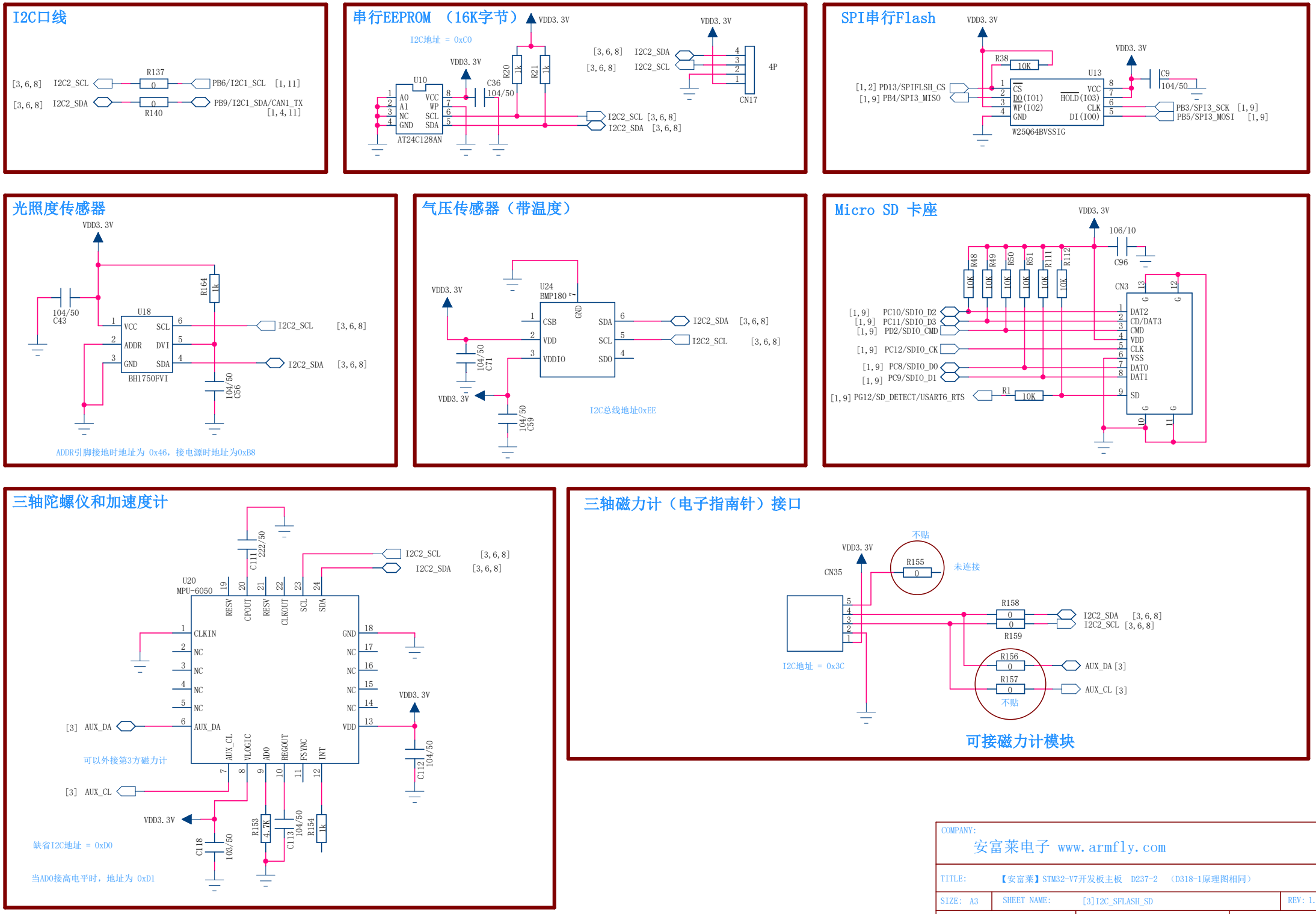
2*13P 2.54mm双排针



2*16P 2.54mm双排针

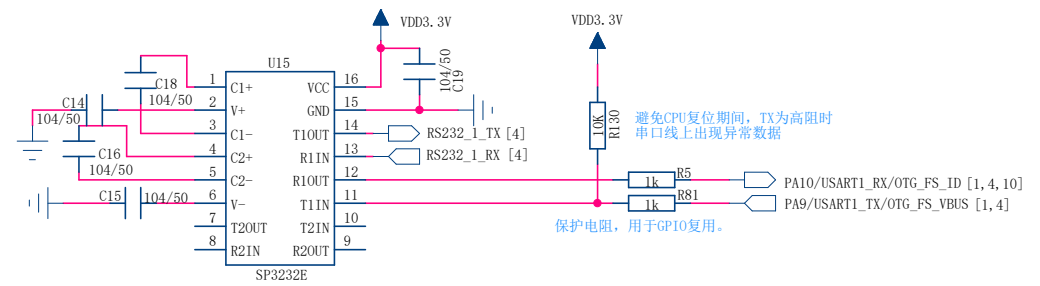


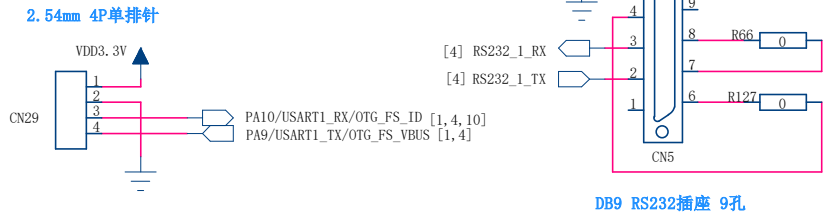
COMPANY: 安富莱电子 www.armfly.com		
TITLE: 【安富莱】STM32-V7开发板主板 D237-2 （D318-1原理图相同）		
SIZE: A3	SHEET NAME: [2]NAND_IOEXPANDOR	REV: 1.0
DRAWN BY: armfly	DATE: 2021-01-04	SHEET: 2 OF 11



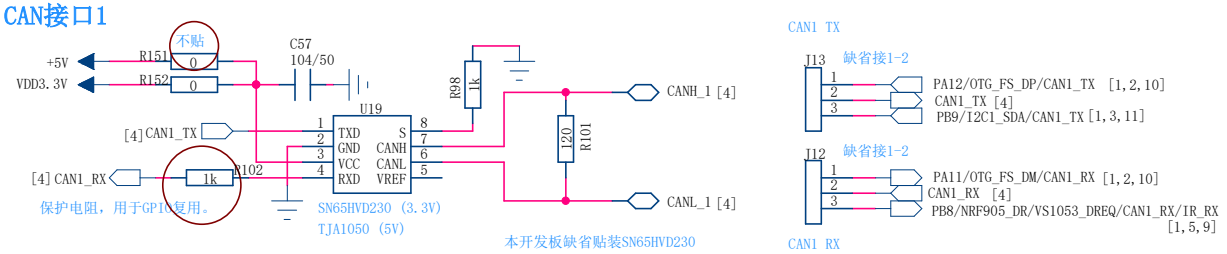
COMPANY: 安富莱电子 www.armfly.com		
TITLE: 【安富莱】STM32-V7开发板主板 D237-2 （D318-1原理图相同）		
SIZE: A3	SHEET NAME: [3]I2C_SFLASH_SD	REV: 1.0
DRAWN BY: armfly	DATE: 2021-01-04	SHEET: 3 OF 11

RS232串口(USART1)

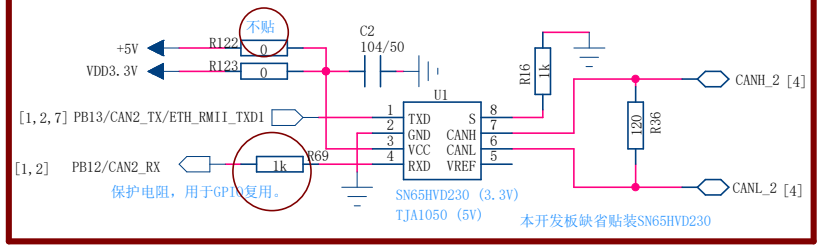




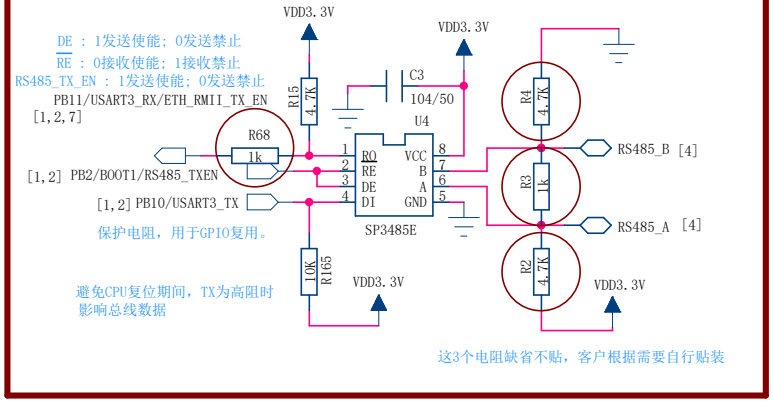
CAN接口1



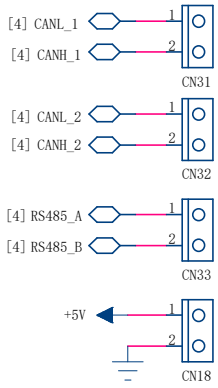
CAN接口2



RS485 (USART3)

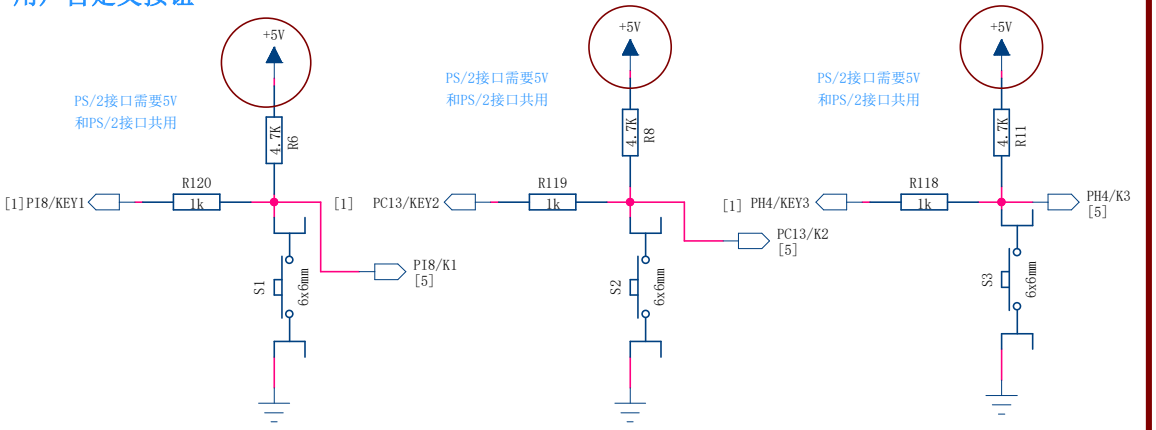


KF128 接线端子



COMPANY: 安富莱电子 www.armfly.com		
TITLE: 【安富莱】STM32-V7开发板主板 D237-2 (D318-1原理图相同)		
SIZE: A3	SHEET NAME: [4]RS232_RS485_CAN	REV: 1.0
DRAWN BY: armfly	DATE: 2021-01-04	SHEET: 4 OF 11

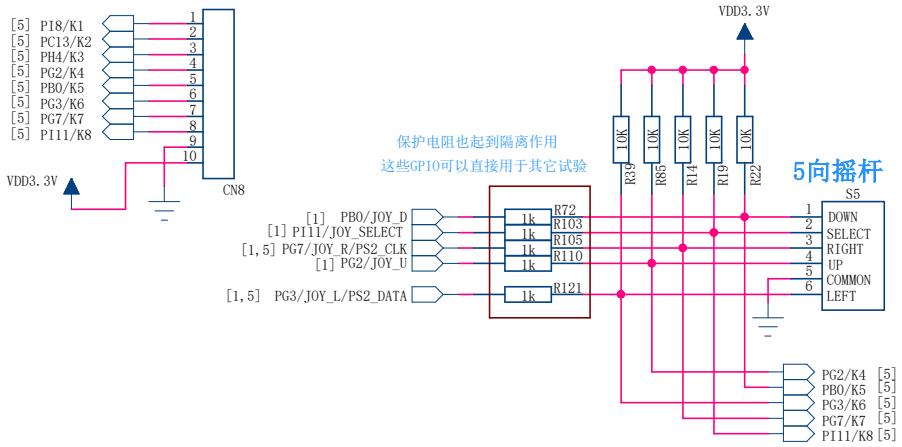
用户自定义按钮



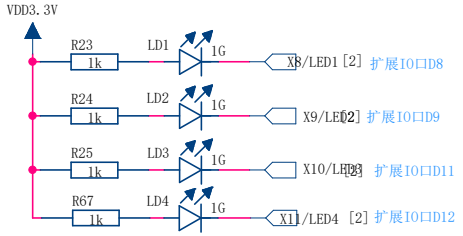
本开发板考虑到IO可能额用作它用，因此IO上并未并联滤波电容
如果做正式产品，建议加上104滤波电容

串联1K电阻的作用：保护GPIO，避免软件错误将IO设置为输出
即使不加滤波，靠软件滤波算法也可正常使用

外接10P按钮排针 2.54mm 10P单排针

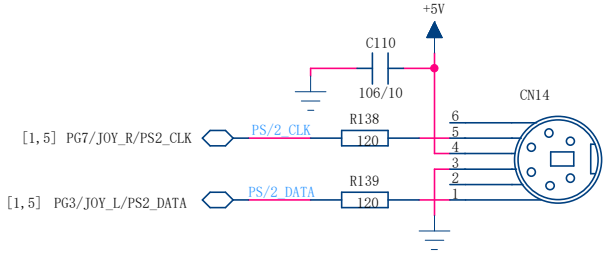


用户自定义LED指示灯



低电平点亮 高电平熄灭

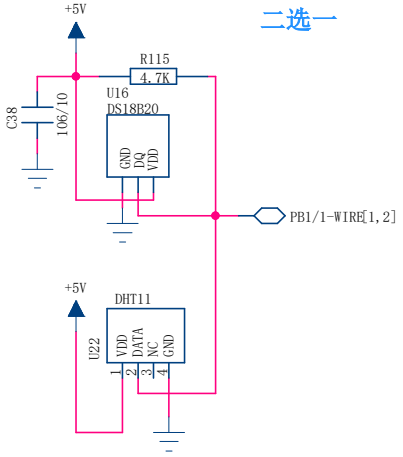
PS/2接口 可连接键盘，鼠标，扫描枪



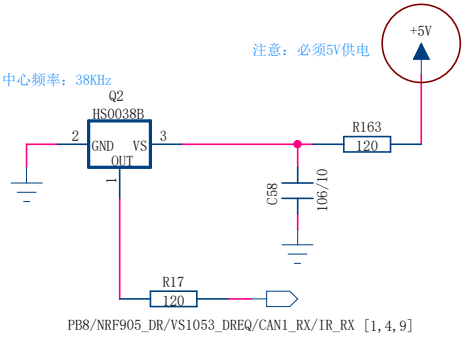
1-WIRE接口

2.54mm 3P圆孔排母

二选一

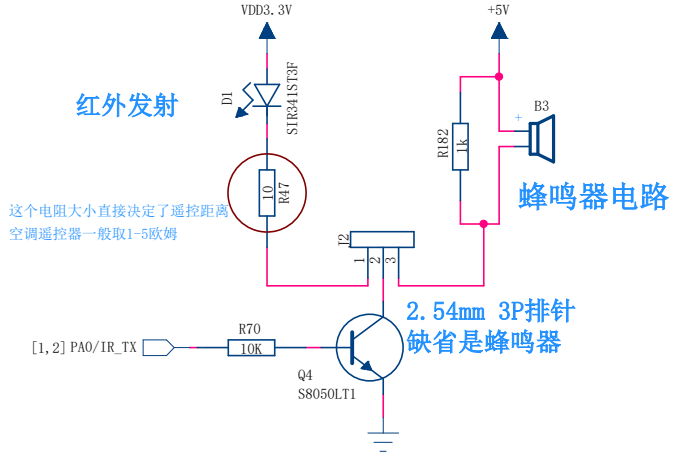


红外接收

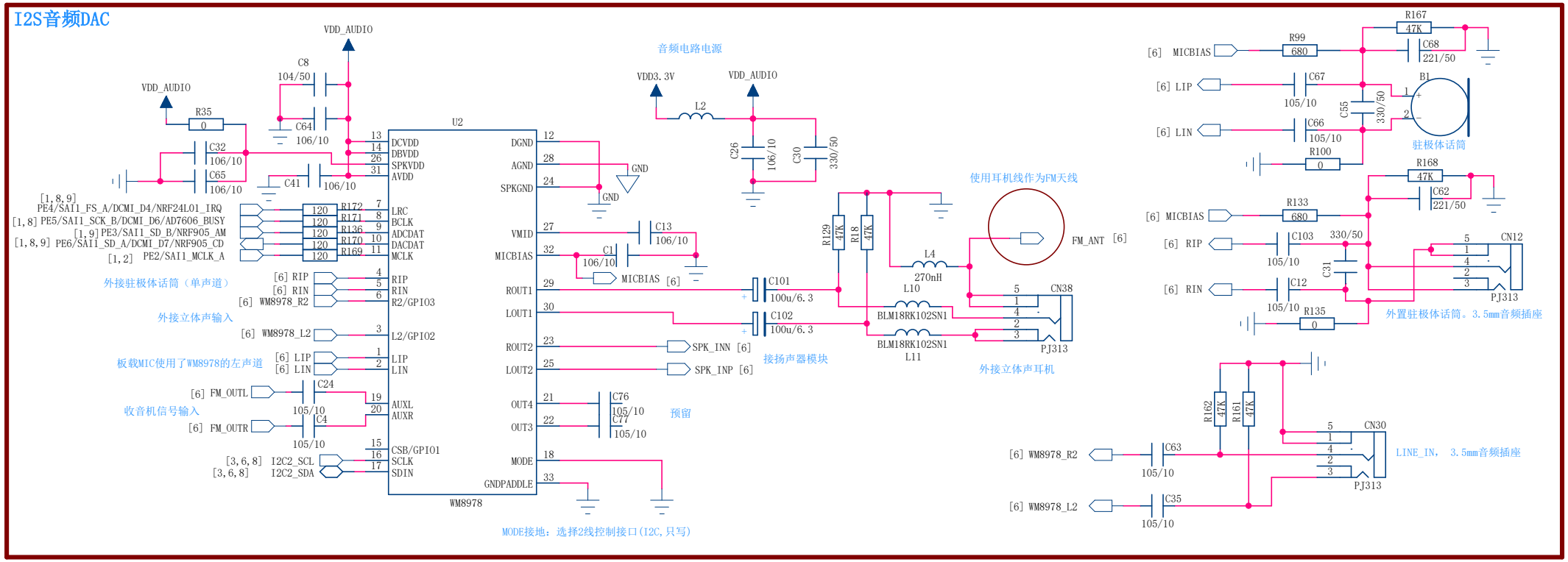


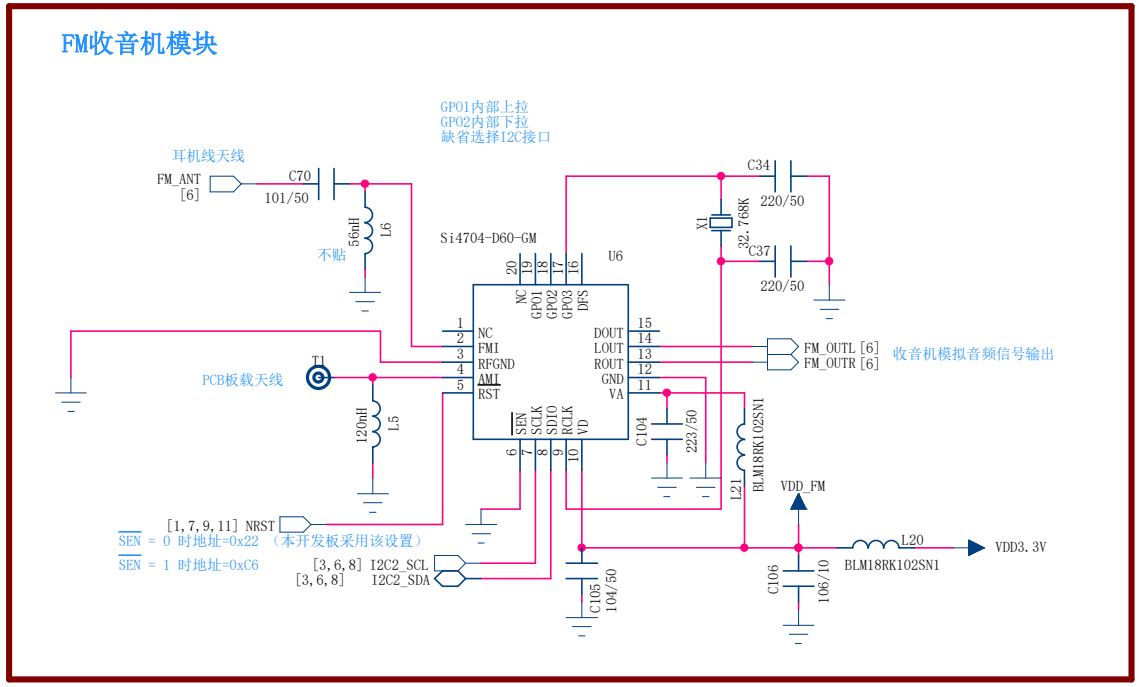
STM32的IO可以兼容5V输入，因此无需电平转换

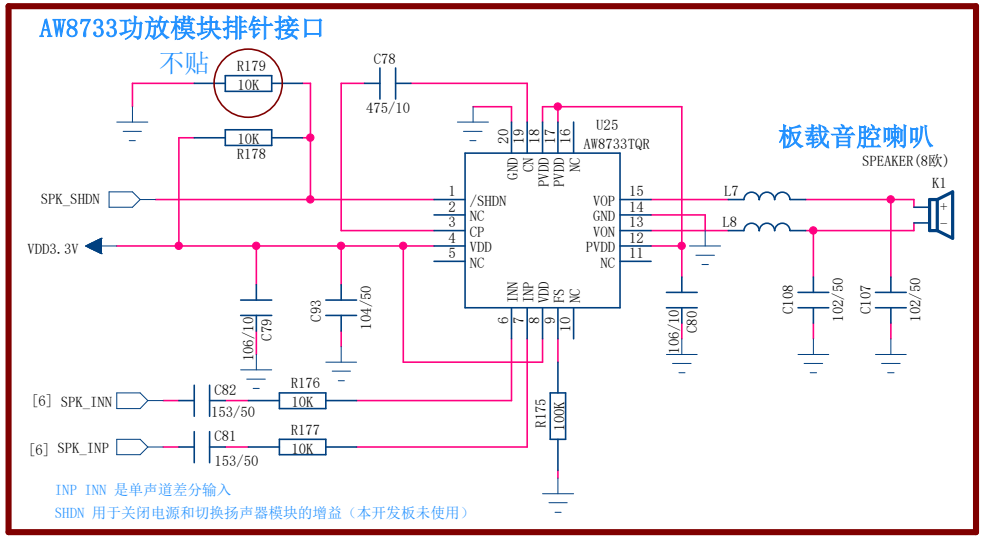
红外发射



COMPANY: 安富莱电子 www.armfly.com			
TITLE: 【安富莱】STM32-V7开发板主板 D237-2 （D318-1原理图相同）			
SIZE: A3	SHEET NAME: [5]KEY_LED_PS2_IR		REV: 1.0
DRAWN BY: armfly	DATE: 2021-01-04	SHEET: 5 OF 11	

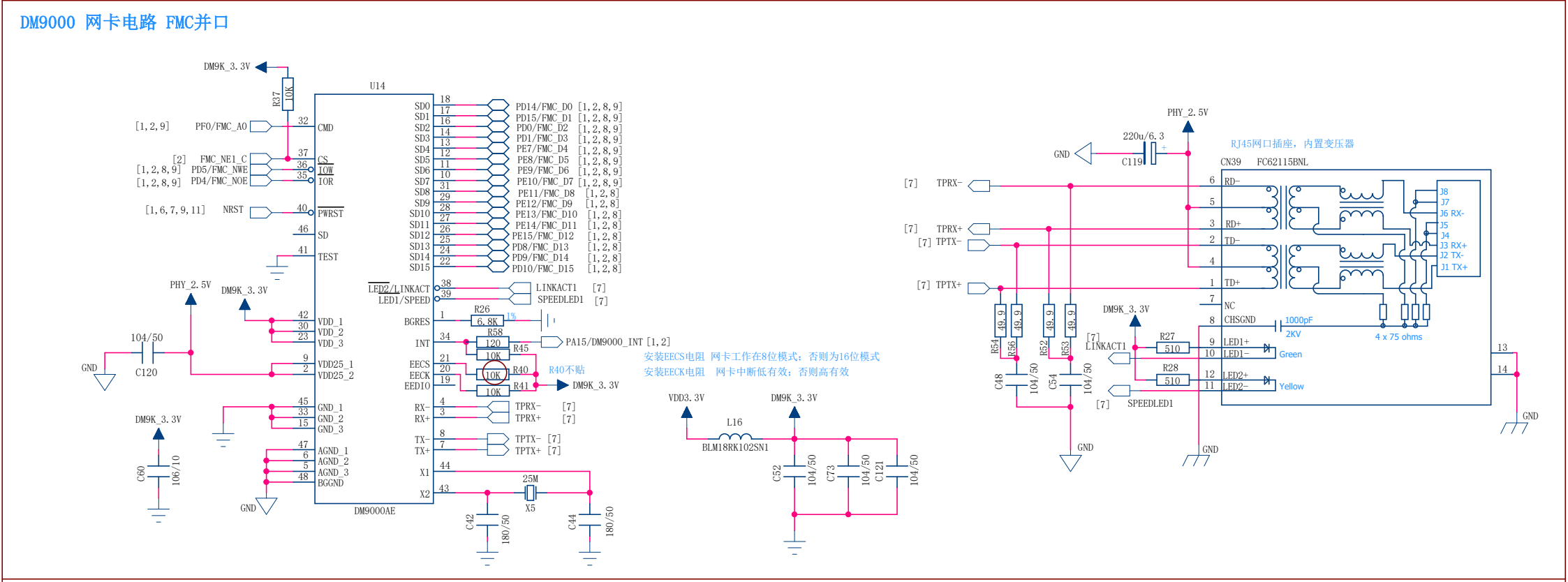




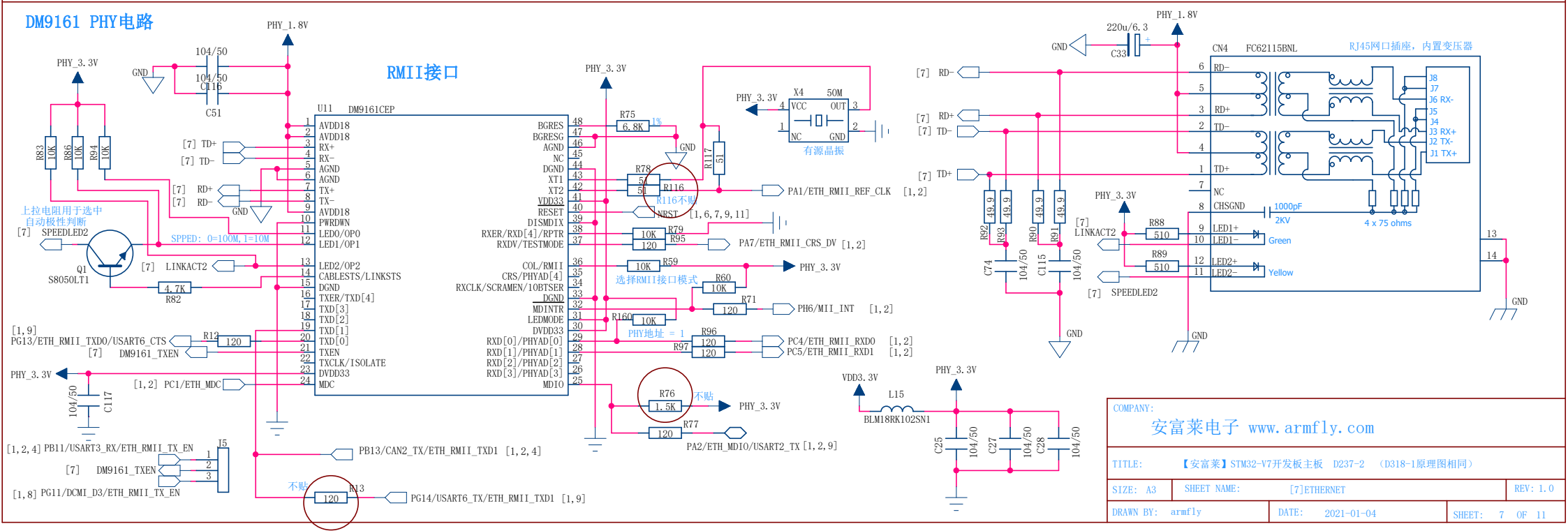


COMPANY: 安富莱电子 www.armfly.com		
TITLE: 【安富莱】STM32-V7开发板主板 D237-2 （D318-1原理图相同）		
SIZE: A3	SHEET NAME: [6]WM8978_FM_SPK	REV: 1.0
DRAWN BY: armfly	DATE: 2021-01-04	SHEET: 6 OF 11

DM9000 网卡电路 FMC并口

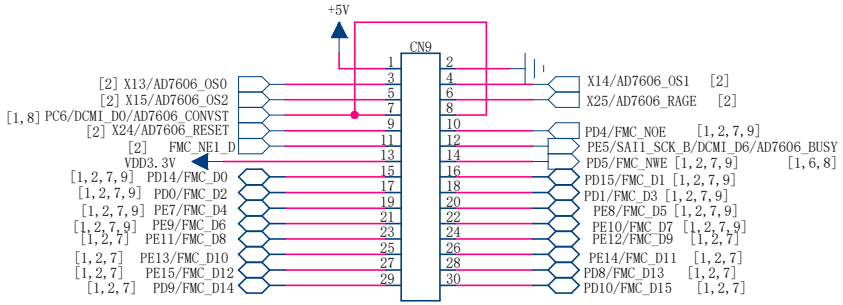


DM9161 PHY电路

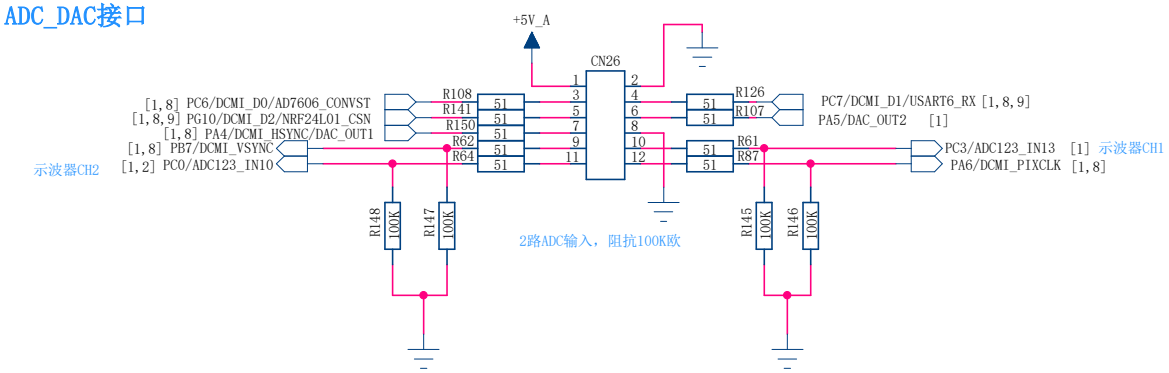


COMPANY: 安富莱电子 www.armfly.com			
TITLE: 【安富莱】STM32-V7开发板主板 D237-2 （D318-1原理图相同）			
SIZE: A3	SHEET NAME: [7]ETHERNET		REV: 1.0
DRAWN BY: armfly	DATE: 2021-01-04	SHEET: 7 OF 11	

AD7606_8通道16位ADC模块接口

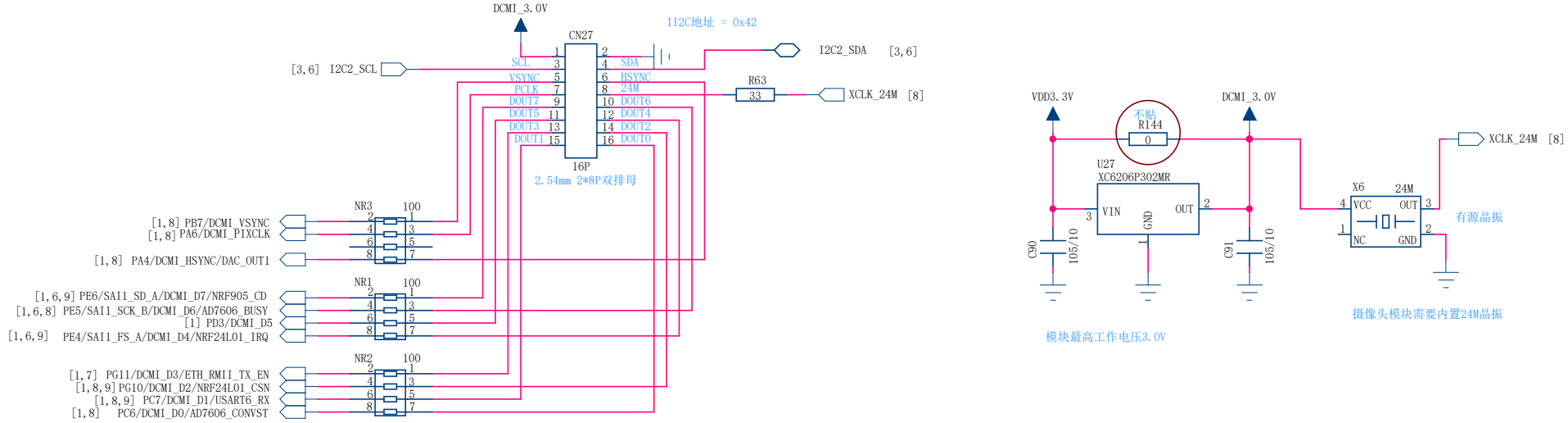


ADC_DAC接口



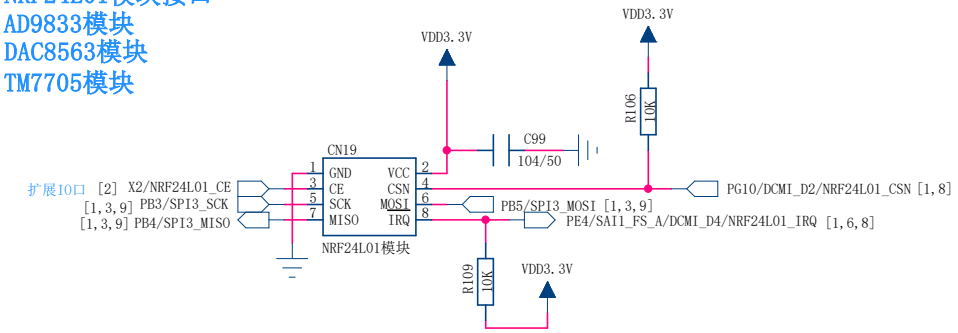
摄像头模块接口

摄像头插上后可以直接使用摄像头
如果使用AD7606，请拔掉摄像头

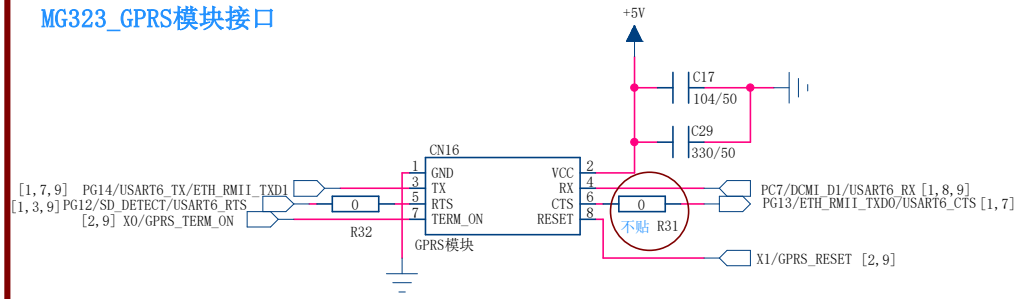


COMPANY: 安富莱电子 www.armfly.com		
TITLE: 【安富莱】STM32-V7开发板主板 D237-2 （D318-1原理图相同）		
SIZE: A3	SHEET NAME: [8]AD7606_CAMERA_ADC_DAC	REV: 1.0
DRAWN BY: armfly	DATE: 2021-01-04	SHEET: 8 OF 11

NRF24L01模块接口
AD9833模块
DAC8563模块
TM7705模块



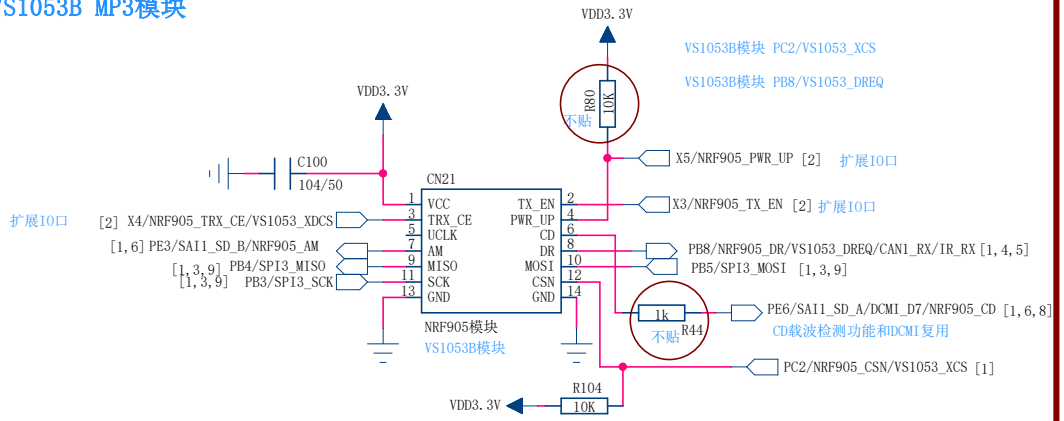
MG323_GPRS模块接口



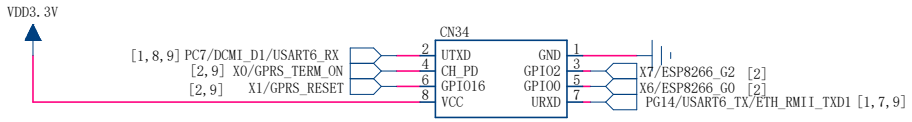
华为MG323_GPRS模块供电电压范围：3.3V - 4.8V
最大峰值电流 2A GPRS模式时平均电流 460mA

由于GPRS电源消耗功率接近USB口供电最大500mA的极限
因此使用GPRS模块时，请使用外部电源供电

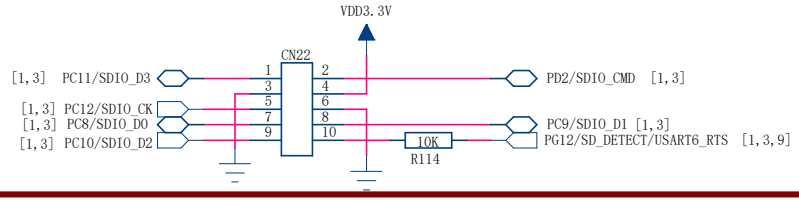
NRF905 通信模块接口
VS1053B MP3模块



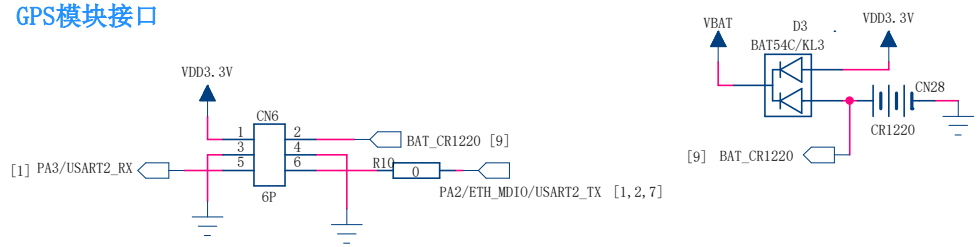
ESP8266串口WIFI接口



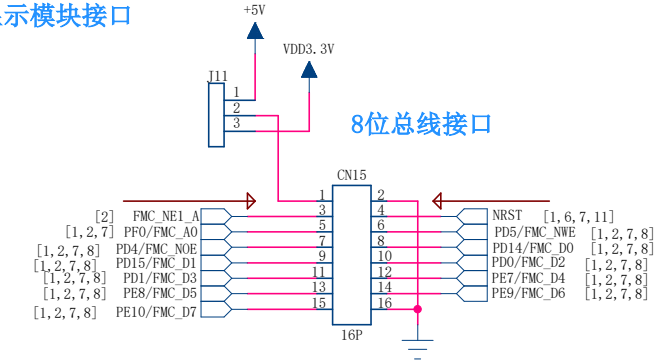
SDIO WIFI模块接口



GPS模块接口



OLED显示模块接口



COMPANY: 安富莱电子 www.armfly.com		
TITLE: 【安富莱】STM32-V7开发板主板 D237-2 （D318-1原理图相同）		
SIZE: A3	SHEET NAME: [9]NRF_GPS_WIFI_GPRS_OLED	REV: 1.0
DRAWN BY: armfly	DATE: 2021-01-04	SHEET: 9 OF 11

外部供电电源电路

输入电压范围：

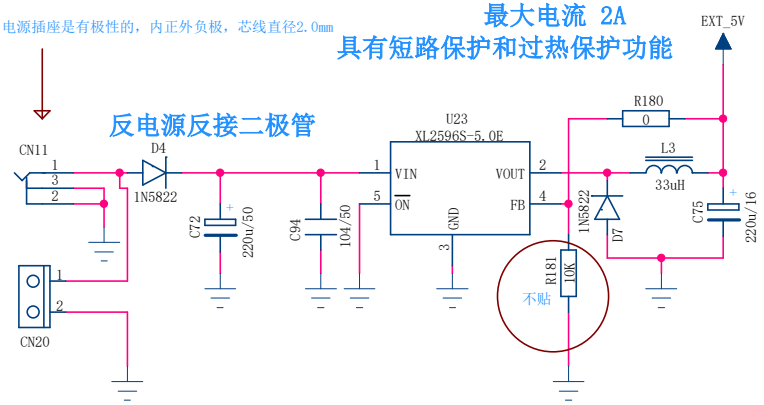
输入电压范围：7~ 40V

最大电流 2A

具有短路保护和过热保护功能

电源插座是有极性的，内正外负极，芯线直径2.0mm

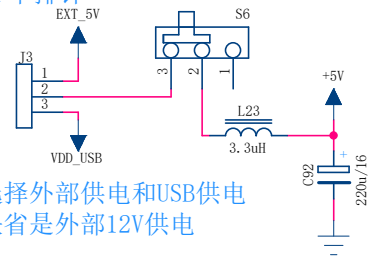
反电源反接二极管



外部供电电源电路

3P单排针

电源开关

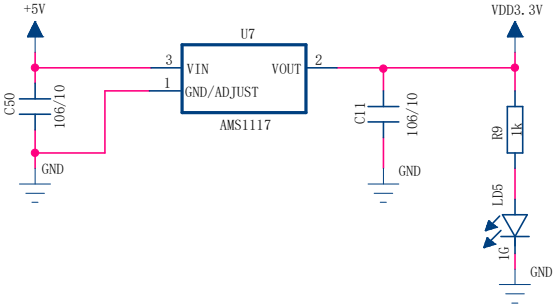


选择外部供电和USB供电

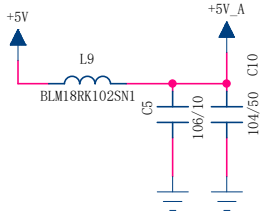
缺省是外部12V供电

外部供电电源电路

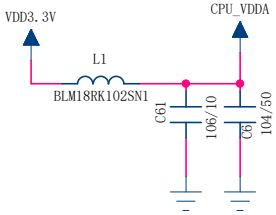
3.3V电源电路（提供给CPU和大部分外设）



ADC/DAC供电电源

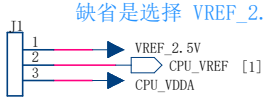


CPU内部模拟部分电源

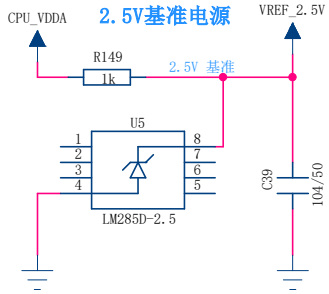


选择CPU内部ADC和DAC参考电压

缺省是选择 VREF_2.5V



2.5V基准电源



USB HOST (USB 2.0 全速12Mbps)

通过数据线连接到PC机

Micro USB插座

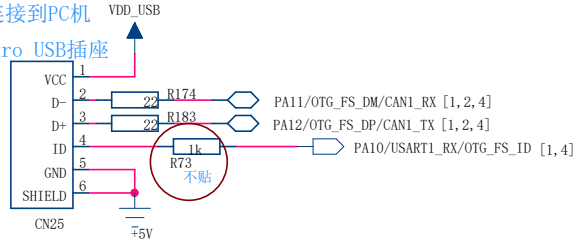
红 +5V

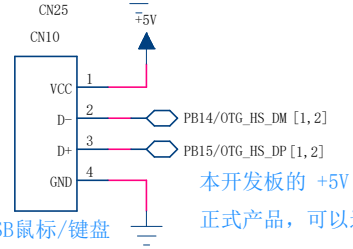
白 D-

绿 D+

ID

黑 GND





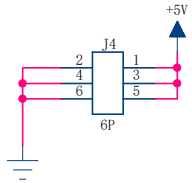
USB-A插座

可插U盘，USB鼠标/键盘

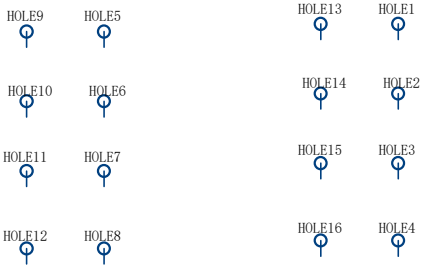
本开发板的 +5V 是带短路和限流保护的

正式产品，可以选择专用功率开关芯片 STMPS2161STR/SOT23-5L

5V电压引出排针

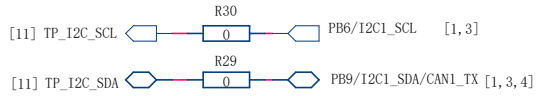


安装孔



COMPANY: 安富莱电子 www.armfly.com		
TITLE: 【安富莱】STM32-V7开发板主板 D237-2 （D318-1原理图相同）		
SIZE: A3	SHEET NAME: [10]POWER_USB	REV: 1.0
DRAWN BY: armfly	DATE: 2021-01-04	SHEET: 10 OF 11

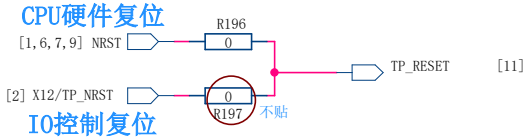
TFT LCD接口（8080-16bit并口）



FPC-0.5-40P下接翻盖FPC插座
可以用于8080总线接口或SPI接口

触摸芯片复位选择

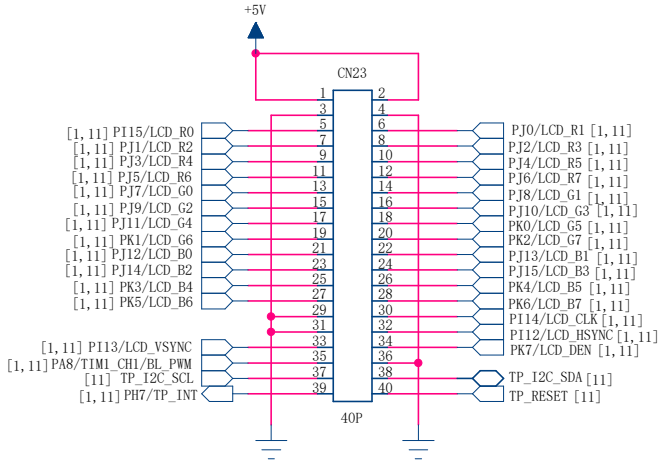
CPU硬件复位



IO控制复位

FPC-0.5-40P插座在核心板上

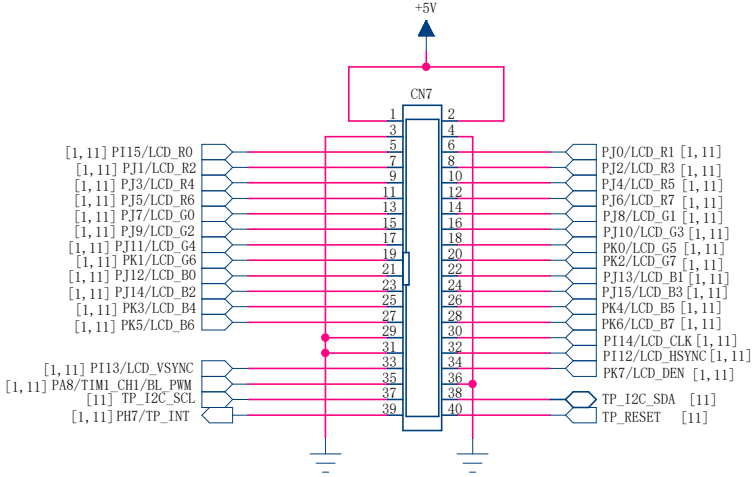
TFT LCD接口（24bit并口）



加高2层双排母（2*20P）

TFT LCD接口（24bit并口）

预留2*20P 2.54mm排针接口 DC3-40P



COMPANY: 安富莱电子 www.armfly.com		
TITLE: 【安富莱】STM32-V7开发板主板 D237-2 （D318-1原理图相同）		
SIZE: A3	SHEET NAME: [11]LCD	REV: 1.0
DRAWN BY: armfly	DATE: 2021-01-04	SHEET: 11 OF 11